

Seth Temple

STATISTICIAN AND APPLIED SCIENTIST

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Education

University of Washington, PhD, Statistics	2024
University of Washington, MS, Statistics	2022
University of Oregon, BS, Mathematics (<i>summa cum laude</i>)	2018

Experience

University of Michigan | Postdoctoral Researcher 09.2024 – Present

- Designed a hypothesis testing method to protect against false discoveries in case-control studies with the Alzheimer's Disease Sequencing Project consortia
- Proposed Bayesian & frequentist infection rate estimators of measles spread w/ partial case data
- Built and taught an AI models workshop on CNN and transformer classifiers for acoustic data

University of Washington | Graduate Student Researcher 09.2019 – 08.2024

- Created a suite of 4 methods to pinpoint and model the effects of advantageous genetic variants
- Developed a fast and scalable algorithm to simulate data for biobank-scale inferences
- Proved central limit theorems in population genetics relevant to downstream analyses
- Performed 5 whole-genome sequence analyses of 3 ancestry groups in TOPMed & UK Biobank to determine regions indicative of selection or technical artifacts

Fred Hutch Cancer Research Center | Graduate Student Researcher 06.2023 – 12.2023

- Designed a new time series method to detect and cluster rapidly evolving SARS-CoV-2 variants
- Collaborated on an international project modeling genomic patterns of synonymous mutations

Los Alamos National Laboratory | Graduate Student Researcher 06.2020 – 10.2020

- Developed and deployed a Bayesian spatial model for mosquito-borne epidemiology
- Implemented spatial ML method to predict and visualize continent-wide mosquito prevalence

Liberty Mutual Insurance | Actuarial Assistant 07.2018 – 08.2019

- Conducted and visualized territory adjustments to ratemaking based on weather and climate risks
- Modernized SQL queries, automated quarterly reserving updates, and analyzed surety claims

University of Oregon | Undergraduate Researcher 01.2018 – 06.2018

- Implemented recurrent neural networks to predict stop words in audio recordings of ESL students

Skills

- Python (8+ years), R (8+ years), SQL and C++ programming
- High-performance computing (slurm)
- Methodology development across disciplines
 - Statistical genetics, genomics, and epidemiology
 - Bayesian inference and hierarchical models
 - Spatial data science and time series
- Deep learning with PyTorch and transformers
- Science communication and collaboration
- Algorithms and software development
- Project planning and execution
- Statistical consulting
- Technical writing

Software

- github.com/sdtemple/isweep: Python package and 5 automated pipelines for genetic analyses (4+ years of continuous software development)
- github.com/sdtemple/btvoccu: R package for Bayesian logistic regression in epidemiology
- github.com/sdtemple/peirrs & github.com/sdtemple/pblas: R packages for estimators in epidemics

Tutorials:

- github.com/sdtemple/zootopia3 & huggingface.co/sdtemple/color-prediction-model: fitting NNs
- github.com/sdtemple/your-first-model: hack-a-thon setup for deep learning on acoustics data
- github.com/sdtemple/how-to-use-ai: using AI coding assistance and agents

Awards

- Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship [UM]
- National Defense Science and Engineering Graduate Fellowship (2% selection rate) [UW]
- Z.W. Birnbaum award for best PhD candidate exam [UW]
- NIH Traineeship in Statistical Genetics [UW]
- Phi Beta Kappa academic honor society [UO]

Teaching

- Lead Instructor for Graduate Statistics Course in Statistical Genetics [UW]
- Teaching Assistant: Intro to Machine Learning, Experimental Design, Applied Regression [UW]
- Directed Reading Program Mentor for 4 students [UW]
- College Math Tutor for 4 years [UO] and K-12 Math Tutor for 1 year [Seattle Public Libraries]

Publications

6 first-author and 5 co-author publications on statistical methods in biological sciences,
3 of which are published in the 8th highest cited Genetics and Genomics journal

1. Temple, S.D., et. al. Multiple-testing corrections in case-control studies using identity-by-descent segments. *The American Journal of Human Genetics* (2026).
2. Temple, S.D., & S.R. Browning. Multiple-testing corrections in selection scans using identity-by-descent segments. *The American Journal of Human Genetics* (2025).
3. Temple, S.D., Waples, R.K., & S.R. Browning. Modeling recent positive selection using identity-by-descent segments. *The American Journal of Human Genetics* (2024).
4. Temple, S.D., & Thompson, E.A. Identity-by-descent segments in large samples. *Theoretical Population Biology* (2025).
5. Temple, S.D., Browning, S.R., & Thompson, E.A. Fast simulation of identity-by-descent segments. *Bulletin of Mathematical Biology* (2025).
6. Temple, S.D., Manore, C.A., & Kaufeld, K.A. Bayesian time-varying occupancy model for West Nile virus in Ontario, Canada. *Stochastic Environmental Research and Risk Assessment* (2022).
7. Temple, S.D., & Terhorst, J. Pair-based estimators of infection and removal rates in partially observed stochastic epidemic models. [Manuscript in progress]

8. Temple, S.D. Statistical Inference Using Identity-by-Descent Segments: Perspectives on Recent Positive Selection. University of Washington (Ph.D. thesis, 2024).
9. Temple, S.D. The Tweedie Index Parameter and Its Estimator: An Introduction with Applications to Actuarial Ratemaking. University of Oregon (Bachelor of Science thesis, 2018).

10. Browning, S.R., Temple, S.D., & Browning, B.L. FLARE2: local ancestry inference with poorly-matched reference panels. *PLoS Genetics* (2026).
11. Haddox, H.K., et. al. Clonal interference and changing selective pressures shape the escape of SARS-CoV-2 from hundreds of antibodies. *Virus Evolution* (2026).
12. Haddox, H.K., Angehrn, G., Sesta, L., Jennings-Shaffer, C., Temple, S.D., et.al. SARS-CoV-2's mutation rate is highly variable between sites and is influenced by sequence context, genomic region, and RNA structure. *Nucleic Acids Research* (2025).
13. Horimoto, A.R.V.R., et. al. Admixture mapping implicates 13q33.3 as ancestry-of-origin locus for Alzheimer disease in Hispanic and Latino populations. *HGG Advances* (2023).
14. Gorris, M.E., Bartlow, A.W., Temple, S.D., et. al. Updated distribution maps of predominant *Culex* mosquitoes across the Americas. *Parasites and Vectors* (2021).